SOLARIS

Solar-Optimized Land Autonomous Recharge & Intelligence System

Critical Design Review

Team 12 | University of Central Florida

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Ryan Markowitz	Computer Engineering
Garrett Fortier	Computer Engineering

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The Team

The Four Engineers Behind it All

Ryan Markowitz

Computer Engineering

Software Lead

Rafael Puig

Electrical Engineering

Power Lead

Nathan Hammond

Electrical Engineering

PCB Lead

Garrett Fortier

Computer Engineering

HW/SW Integration Lead

Motivation & Background

Why Mobile Solar?

The Problem

- Static Solar Panels can't autonomously adapt to shaded environments
- Typical basic Solar Panels systems don't allow direct USB-A device charging.
- Typical Solar Panels don't allow users to wirelessly track harvested energy statistics.
- Usual solar platforms are not adapted to a multitude of environments.

Our Angle

- SOLARIS moves when it detects it's in shade for optimal power gains over time
- The robot has direct USB-A charging capabilities to interface with a variety of devices.
- Send telemetry information to end user via a phone app connected through Bluetooth Low Energy

OWNER : RYAN

Goals & Objectives

From Basic to Advanced

BASIC

- Rotate the solar panel about two axes to orient itself in peak sunlight conditions
- Create a phone app to serve as the primary means of communication between the user and the robot
- Measure and report information to the end user such as distance traveled, net energy, state of charge, etc
- Robot is capable of autonomously moving out of shade on flat ground into a highly luminous area
- Ensure the system can operate autonomously without external power or physical user input after deployment for at least 12 hours.

ADVANCED

- Energy gained from moving the robot must be greater than energy used during runtime
- Robot accurately avoids collision autonomously
- Allow users to manually select between three different modes through phone app
  - Stationary: the robot will not move, however it will still move the solar panel
  - Autonomous: full control is given to the robot
  - Manual: receive throttle and steering information from user to move around
- Integrate motion sensing to quantify distance traveled during movement operation.
- Design the electronics and firmware architecture with modularity.

OWNER : GARRETT

Engineering Specifications

Overall System

Specification	Description	Target Value and Unit(s)
Energy for User	After a sunny, optimal day, the system shall provide the end user with at least 15 watt-hours (Wh) of usable energy from the battery.	≥15 Wh/day
Max Forward Robot Speed	Normal forward robot speed.	0.3 m/s
Max Reverse Robot Speed	Normal reverse robot speed.	0.3 m/s
Solar Panel Tilt	Tilt axis range of motion	30 Degrees
Solar Panel Rotation	Pan axis range of motion	360 Degrees
Weight	Total weight of robot	<50 lbs.

Component System

Component	Parameter	Target Value and Unit(s)
Solar Panel	Power	≥15 Wh/day
Battery Capacity	Maximum Battery Capacity	256 Wh
Bluetooth Low Energy (BLE)	Wireless Communication Range	1-6 meters
Geared Brushed DC Motor	Torque	18.14 Nm, 24.5 Nm

OWNER : GARRETT

Five Subsystems

Each with their own deep-dive!

Control

ESP32-S3 brain + firmware

Drivetrain

4 motors, skid steering

Solar Tracking

2-axis arm + phototransistor array

Power

Panel → MPPT → LiFePO4 → rails

Communications

BLE + companion app

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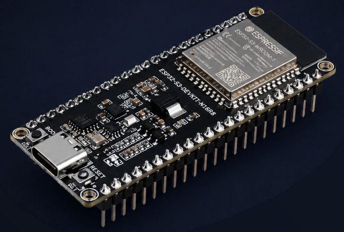
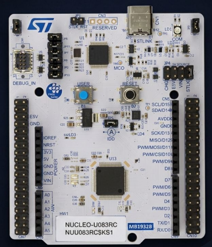
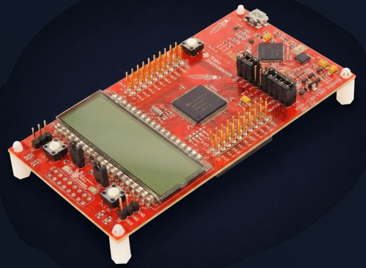
Control Unit Selection

Comparing MCU candidates







Microcontroller	Architecture	Bluetooth /Wi-Fi	GPIO Count	ADC	Development Ecosystem	Cost Per Unit
STM32U0	Cortex-M0+	None	~29	1x 12-bit, 16 channels	STM32Cube IDE, Arduino, PlatformIO	$4.96
MSP430FR6989	16-bit RISC	None	~83	1x 12-bit, 16 channels	TI Code Composer	$12.49
ESP32-S3	Xtensa LX7 Dual-Core	BLE/Wi-Fi 4 (802.11 b/g/n)	~45	2x 12-bit, 20 channels	ESP-IDF, Arduino, PlatformIO	$6.76
ESP32-P4	RISC-V Dual-Core	None	~54	2x 12-bit, 10 channels	ESP-IDF, Arduino	$59.74*

SELECTION

ESP32-S3

- **Integrated Wi-Fi + BLE** — integrated wireless connectivity not requiring an external radio
- **GPIO Density** — flexible pin matrix for routing and peripheral mapping
- **Strong Community** — widely used with mature software ecosystem for quicker prototyping time
- **Price To Performance** — Dual core processor utilizing RTOS with rich peripheral set

OWNER : GARRETT

*at time of creation access to chipset is very limited and only could be found on devboards

Drive Motor Technology Selection

Brushed vs Brushless vs AC

Motor Type	Control Complexity	Motor Driver	Efficiency	Cost	Battery Compatibility	Low Speed Torque
AC	High	VFD/AC Drive	High	High	Poor	Requires Additional Control Methods
Brushless AC	High	Electronic Speed Controller (ESC)	High	Moderate	Excellent	Good
Brushed DC	Low	H-Bridge	Moderate	Low	Excellent	Good

SELECTION

Brushed DC Motor

- Lowest Cost
- Simple Control
- Great Torque Characteristics

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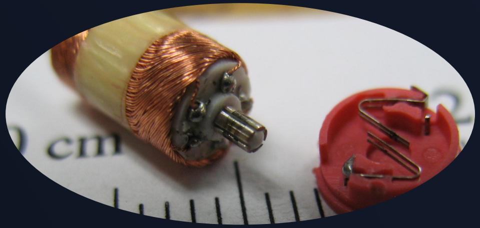
Drive Motor Selection

DC brushed drive motor selection



Motor Model	Voltage	Rated Torque	Rated Speed	Gear Ratio	Encoder	Cost
StepperOnline F5D60-12GN-30S-5 GN50K	12V	6.27 Nm	60 RPM	50:1	No	$52
Duma Dynamics 13PPR Encoder 60RPM DC Gearmotor	12V	5 Nm	60 RPM	104:1	Yes	$59
goBILDA 5204 Series Yellow Jacket Planetary Gear Motor	12V	18.14 Nm	60 RPM	139:1	Yes	$57

SELECTIONS

goBILDA 5204 Yellow Jacket

- Meets Documentation Standards
- Exceeds Torque Requirements
- Built-in Encoder
- High Gear Ratio

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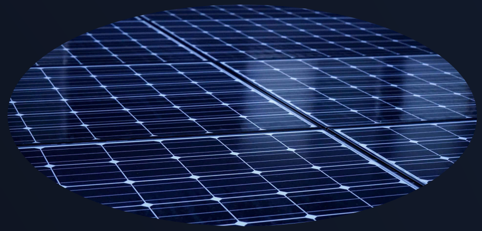
Solar Panel Selection

Analyzing different monocrystalline solar panels



20W 12V Solar Panel Model	Temperature Coefficient of Pmax	Size	Weight	Power Density (mW/in^2)	Cost
ACOPOWER HY020-12M	-0.45%/°C	13.6 × 18.5 × 0.8 in	6.2lbs	79.5	$39.50
Newpowa NPA20S-12J	-0.36%/°C	15.6 × 12 × 0.9 in	3.16lbs	106.8	$39.99
SLD TECH ST-20P-12	-0.5%/°C	15.4 × 14.2 × 1.2 in	4lbs	91.5	$63.80
Ameresco C1D2-420J	-0.45%/°C	16.7 × 19.8 × 2 in	6lbs	60.5	$167.70

OWNER : RAFAEL

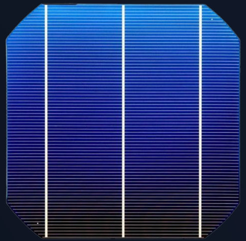
SELECTION

Solar Power Selection

- **Highest Power Density -** outstanding power generated per area.
- **Affordability -** Very low cost solar panel.
- **Grade A Panel -** Free from defects.

Solar Panel Technology Selection

Monocrystalline vs Polycrystalline vs Thin-Film



Solar Panel Technology	Efficiency	Weight	Cost	Temperature Performance	Degradation Rate
Monocrystalline Silicon	Excellent	Good	High	Good	Low
Polycrystalline Silicon	Good	Good	Moderate	Moderate	Moderate
Thin-Film Silicon	Low	Excellent	Low	Best	High

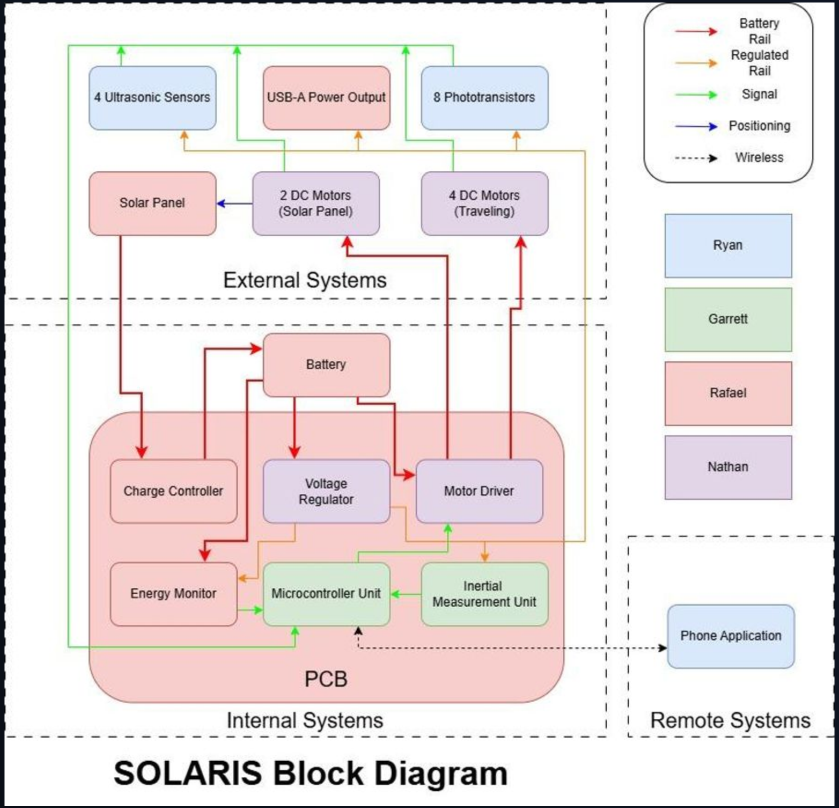
SELECTION

Solar Power Technology

- **Highest Efficiency -** outstanding power generated per area.
- **Manageable weight -** decent mass per power generated.
- **Durable -** High longevity rating.

OWNER : RAFAEL

Hardware Block Diagram



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Light Sensor Technology Selection

Photodiodes vs Phototransistors vs LDR vs I2C IC

Sensor Type	Ease of Implementation	Detect Sun vs. Shade	Multi-Sensor Scalability (4-8 sensors)	Detect Small Directional Differences	Cost
Photoresistors (LDR)	Easy	Yes	No issues	Moderate (works well with shading geometry)	Low
Phototransistor	Easy	Yes	No issues	Good (more sensitive than LDRs, but still benefit from shading geometry)	Low
Photodiode	Medium / Hard	Yes	No issues	Very good (with an amplifier. Raw signal is small)	Low
Digital Light Sensor (LUX, I2C)	Medium	Yes	Potential Addressing issues	Excellent. Gives exact LUX values. In terms of accuracy it is the best.	Medium

SELECTION

Phototransistors

- **Built-in gain** — strong analog signal from just one load resistor, no amplifier stage
- **40% stronger signal than LDRs** — per published tracking study; better directional sensitivity
- **No I²C address conflicts** — scales cleanly to 4+ sensors without any addressing issues
- **Direct 3.3 V ADC read** — trivial software, just sample and compare
- **Relative tracking, not absolute lux** — best balance of sensitivity, simplicity, and cost

OWNER : RYAN

Light Sensor Selection

Narrowing down the phototransistor candidates

Sensor	Collector Light Current	Viewing Angle	Wavelength Range	Dark Current	Max Operating Voltage	Cost Per Unit
VTT9812FH	0.1 mA	56°	450-700 nm	100 nA	30 V	$0.57
VTT9814FH	0.1 mA	50°	450-700 nm	50 nA	30 V	$0.63
TEPT5600	20 mA	20°	440-800 nm	50 nA	6 V	$0.58
TEPT4400	20 mA	30°	440-800 nm	50 nA	6 V	$0.40

SELECTION

TEPT5600

- **Narrow 20° viewing angle** — sharpest directional contrast between bright and shaded sides
- **Visible response (440–800 nm)** — tracks sunlight without IR skewing the readings
- **6 V rating** — fully compatible with the 3.3 V system
- **Through-hole on breakout boards** — flexible, rugged external mounting on the chassis
- **Saturation managed** — high sensitivity tuned out with load-resistor selection
- **Others ruled out** — VTT parts too wide-angle (washes out contrast); TEPT4400 hard to find in stock

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Charge Controller Technology Selection

MPPT vs PWM

Solar Panel Charge Controller Technology	Power Efficiency	Daily Energy Capture	Safety	Complexity	Cost
Multi Power Point Tracker	95-99%	High	High	High	High
Pulse Width Modulation	70-80%	Moderate	Moderate	Moderate	Moderate

SELECTION

Charge Controller Technology Selection

- **Highest Efficiency -** Wastes least amount of potential power from sunlight.
- **Max Energy Capture -** Maximizes power over day.
- **Safety -** More stable operation.

OWNER : RAFAEL

Charge Controller Selection

Selecting a charge controller

Solar Panel Charge Controller Part	Electrical Efficiency	Operating Power Consumption	Key Features	Weight	Cost
GV-5-LI-14.2V	94-99.85%	0.19 mA	Fixed charge algorithm, 10-year warranty, IP40, 6-position terminal block connection	0.176lbs	$99
GV-5-MOD-LFP	94-99.85%	0.15 mA	Fixed charge algorithm, 10-year warranty, 24-pin connection	0.062lbs	$80
Victron SmartSolar MPPT 75/10	≤98%	20 mA	Built-in Bluetooth, programmable charge algorithm, 5-year warranty, IP22, screw terminal connection	1.102lbs	$73
RNG-CTRL-RVR20-US	≥95%	≤100 mA	Optional Bluetooth add-on, programmable charge algorithm, 3-year warranty, IP32, screw terminal connection	3.1lbs	$116

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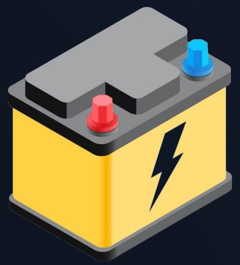
SELECTION

Charge Controller Selection

- **Low Operating Power -** Minimizes wasted power through controller.
- **Low Weight -** Low added mass to system.
- **Environmental Protection-** Durable against nature.

Battery Technology Selection

Lithium Ion vs Lithium Iron Phosphate vs Lead-acid



Battery Technology	Specific Energy (Wh/kg)	Depth Of Discharge	Pros	Cons	Cost ($/kWh)
Lithium Ion	~150-250	80-90%	Very high energy density, good power capability	Moderate safety risk if abused, requires BMS, shorter cycle life than LiFePO4, thermal runaway possible	150-300$
Lithium Iron Phosphate	~90-120	80-100%	Excellent safety and thermal stability, very long cycle life (2000–5000+), good for deep cycling and solar storage	Lower energy density (heavier than NMC), higher upfront cost than lead-acid	200-400$
Lead-acid	~30-50	50%	Low upfront cost, simple charging, highly recyclable	Very heavy, short cycle life, poor deep-discharge capability, lower efficiency	60-150$

SELECTION

Lithium Iron Phosphate

- **High Depth of Discharge -** The rated energy storage of the battery is able to be fully used.
- **Safety -** Significantly more stable than Lithium Ion batteries.
- **High Specific Energy -** Battery has a lower weight per energy capacity than Lead-acid.

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Battery Selection

Choosing the right battery

Battery Part	Rated Capacity	Operating Temperature	Size	Weight	Cost
Dr.Prepare DBT1220LFP	20Ah (256Wh)	-10-45°C	7.13 × 3.03 × 6.61 in	5.95lbs	$79.99
Bioenno Power BLF-1220AS	20Ah (240Wh)	-10-60°C	7.17 × 3.03 × 6.73 in	5.73lbs	$204.99
Renogy RBT1220LFP-TM-G1	20Ah (256Wh)	-20-60°C	7.13 × 3.03 × 6.61 in	5.95lbs	$89.99
Ultralife URB12200	21.6Ah (276Wh)	-20-60°C	7.12 × 3.03 × 6.57 in	5.97lbs	$285.13

SELECTION

Battery Selection

- **Large Temperature Range -** able to continuously operate up to 60°C
- **Low Cost -** Affordable for project budget constraints.
- **Reliable Vendor-** Renogy is well-known solar system provider, and provides product warranty.

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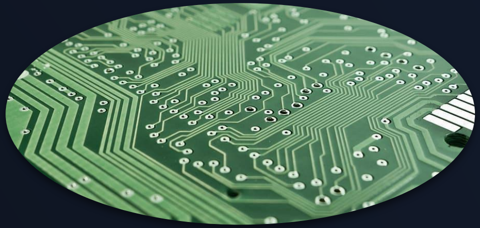
Voltage Regulation Technology Selection



Justifying Regulator Selection

Voltage Regulator Technology	Power Efficiency	Daily Energy Capture	Safety	Complexity	Cost
Switching	High Efficiency, Lower Power Dissipation for Large Voltage Drops	Maximizes Battery Energy, Reduced Thermal Loss	Lower Heat Generation Reduces Thermal Stress	Moderate. Requires More Careful PCB Layout and More Components	Moderate. More Components Means a Higher Cost
Linear	Efficiency is Linear Therefore Larger Voltage Drops Result in Heat Dissipation	Excess Voltage Dissipated = Reduced Usable Battery	Lower Noise and EMI, but Needs Thermal Management at High Currents	Low. Less Components, Simpler PCB Layout	Low. Fewer Components and Simple Architecture

SELECTION

Switching Regulator

- Higher Efficiency
- Maximizes Battery Life
- Lower Thermal Stresses

OWNER : NATHAN

Voltage Regulation Selection

Regulator Part Selection

Voltage Regulator	Conversion Efficiency	Output Channels	Key Features	Max Current Per Channel	Quiescent Current	Cost
LM2679	84%	1	High Current; Current Limit Adjustable	5 A	~7 mA	$7.42
LM2576	77%	1	Robust; Industry Standard	3 A	~5 mA	$1.42
LM2596	73%	1	Common; Easy to Implement	3 A	~5 mA	$2.34
LT8650S	93%	2	Dual Outputs; Synchronous; Silent Switcher (Low EMI)	4 A	6.2 uA	$18.31
LTC3633	93%	2	Dual outputs; Compact	3 A	2 mA	$12.69

SELECTION

LM2679

- Easy to Implement
- Proven To Be Reliable
- Can use Smaller Passive Components
- Can Use Same Regulator for 3.3 V and 5 V

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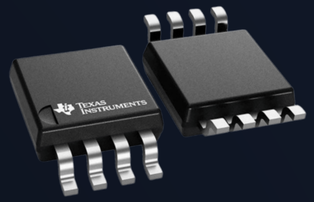
Energy Monitoring Selection



Identifying the best IC

Energy Monitoring Part	Current Accuracy	Bus Voltage Range	Accessible Measurements	Cost of One IC
LTC2944	±1%	3.6-60 V	Battery Voltage, Current, Die Temperature, Accumulated Charge	$12.82
INA226	±0.02-0.1%	0-36 V	Shunt Voltage, Battery Voltage, Current, Power	$2.99
INA228	±0.01-0.05%	-0.3-85 V	Shunt Voltage, Battery Voltage, Die Temperature, Current, Power, Accumulated Energy, Accumulated Charge	$5.27

SELECTION

Energy Monitoring Selection

- **Direct Energy Calculation-** Simplifies external computation needed by having on chip energy registers.
- **Highly Accurate -** provides high fidelity readings of energy consumed and produced by system.
- **Affordability -** IC is reasonably priced.

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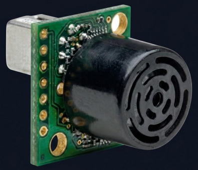
Ultrasonic Sensor Selection

Picking the Right Crash Avoidance Sensor


Sensor	Operating Voltage	Typical Current Draw	Beam Angle	Range	Minimum Time Between Readings	Communication Interface	Cost Per Unit
MB1020-000	2.5-5.5 V	2 mA	42°	.15-6.5 m	50 ms	UART + ADC	$19.95
MB1232-000	3-5.5 V	Not Specified	44°	.2 - 7.7m	25 ms	I2C	$37.44
MB1013-000	2.5-5.5 V	2.5-3.1 mA	44°	.3-5 m	100 ms	UART + ADC	$31.19

SELECTION

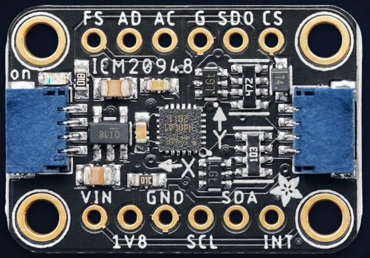
MB1020-000

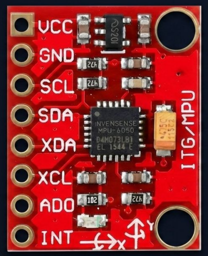
- **42° beam angle** — sits right in the 40–45° target, balancing coverage against range and false readings
- **6.5 m range** — comfortably clears the 5 m detection requirement we set out for SOLARIS
- **Lowest current draw (2 mA)** — the lightest power load of the candidates, fitting the solar budget
- **Sensor chaining supported** — sequential triggering kills crosstalk without complex firmware timing logic
- **Cheapest at $19.95** — decisive when four units are needed across the chassis

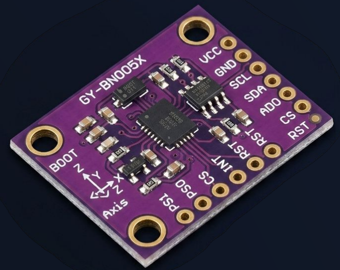
OWNER : RYAN

IMU Selection

(Development Kits)







Specifications	BNO-085	MPU-6050	ICM-20948
Degrees of Freedom	9-DoF	6-DoF	9-DoF
Onboard Processing	Full Sensor Fusion (ARM Cortex-M0)	Basic Digital Motion Processor (DMP)	Advanced DMP (to offload calculations)
Power Profile	Low Power Modes	Sleep Register	Low Power Mode
Stability	High (Absolute Orientation)	Medium (No Magnetometer)	Medium-High (Relative Compass)
Communication Protocol	(I2C, SPI, UART)*	I2C	I2C & SPI
Cost	$24.95	$12.95	$19.95

SELECTION

ICM-20948

- **9-DoF sensing** — integrates accelerometer, gyroscope, and magnetometer all in one
- **I2C and SPI** — flexible interface options for prototyping and for bus combination
- **Cost Effective** — delivers 9-DoF without the expense of onboard fusion processors
- **Ease of Use** — Simple I2C communication bus with no external wrapper needed
- **Low Power** — suited for battery operated solar-powered embedded system

OWNER : GARRETT

*communication must be put into a sh2 protocol wrapper via a dedicated library

App & Backend Stack

Cross-platform mobile

Frontend	React Native	*vs SwiftUI, Flutter*
BLE Client	React-native-ble-plx	
Backend	FastAPI (Python)	*vs Django, Spring Boot, Express*
Database	PostgreSQL on Supabase	
Auth	Clerk	

OWNER : RYAN

Wireless Protocol Selection

Why Bluetooth Low Energy (BLE)

	Wi-Fi	Bluetooth Standard	Bluetooth Low Energy	LoRaWAN	Zigbee
Range (m)	20-100	Up to 30	Up to 50	10,000	1,000
Data Rate	Up to 46 Gbps	1-3 Mbps	.5-1 Mbps	27Kbps	500 Kbps
Power Consumption	Highest	Medium-Low	Low	Lowest	Low
Frequency Bands	2.4GHz/5Ghz/6GHz	2.4GHz	2.4GHz	915MHz	915MHz/2.4GHz
Phone Compatibility	Yes	Yes	Yes	No: separate hub required	ZigBee direct: Use Bluetooth to connecto to Zigbee IOT devices

SELECTION

Bluetooth Low Energy (BLE)

- **Native phone connectivity** — pairs straight to the app with no router, hub, or gateway required
- **Low power** — lighter power draw than some of the other candidates
- **Right-sized bandwidth** — comfortably handles low-rate telemetry and short control commands without wasted overhead
- **Easy app integration** — well-supported in mobile development, simplifying the companion app
- **Range tradeoff accepted** — shorter reach than Wi-Fi, but fine for a locally controlled, nearby user

OWNER : RYAN

Embedded Stack

Firmware Architecture: ESP-IDF

- Native FreeRTOS Integration: Bypasses Arduino abstraction layers and allows for real time task scheduling across the ESP32-S3’s dual cores.
- Unrestricted Peripheral Access: Direct access to advanced ESP32-S3 hardware (Motor Control PWM, vector instructions) without overhead.
- Official Toolchain: Unlike PlatformIO ESP-IDF guarantees zero-day access to Espressif patches, bug fixes, and other updates.

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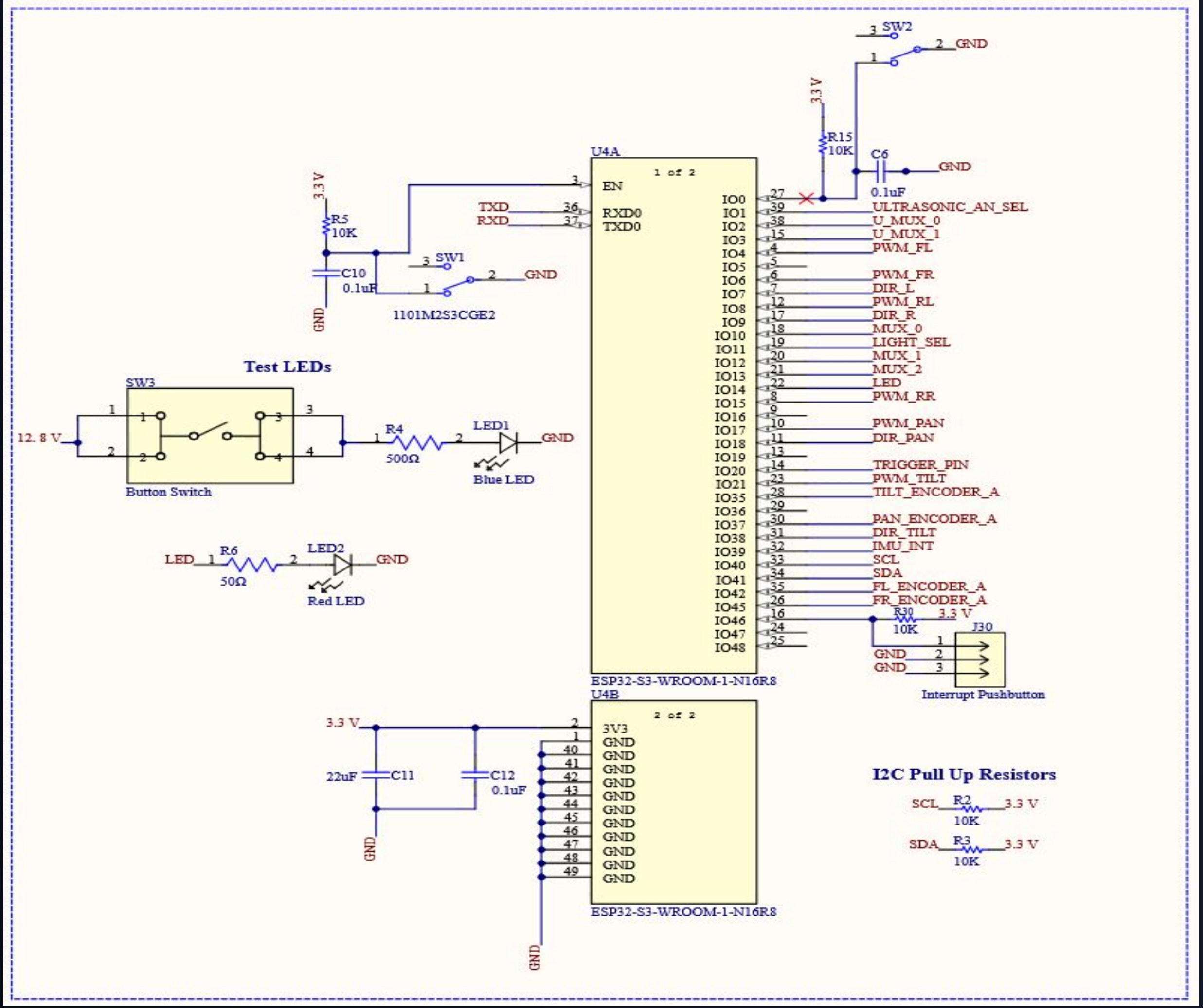
Main PCB Schematic

MCU + Sensors + Power



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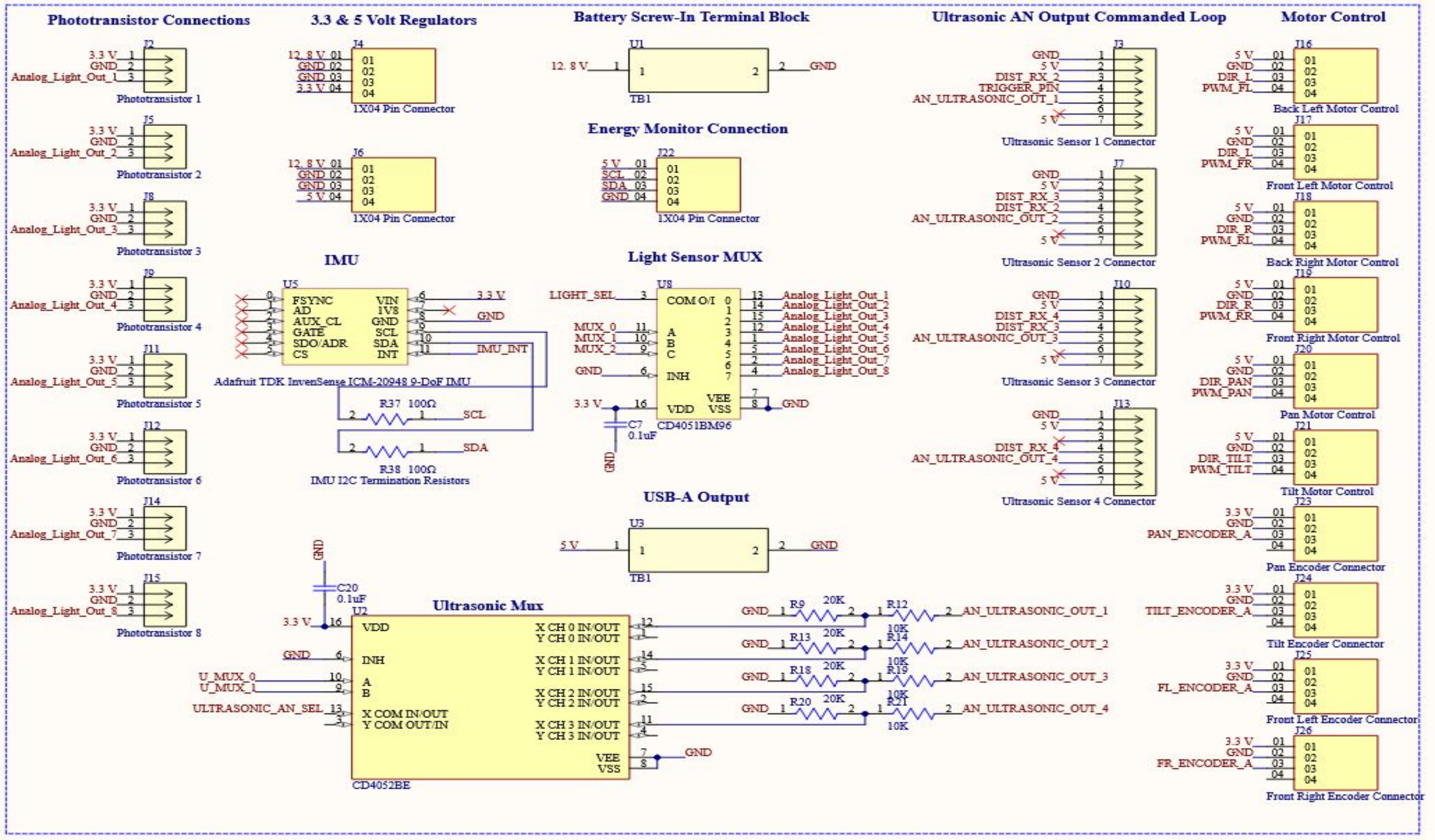
Sensors and Sensor Connections

Sensors



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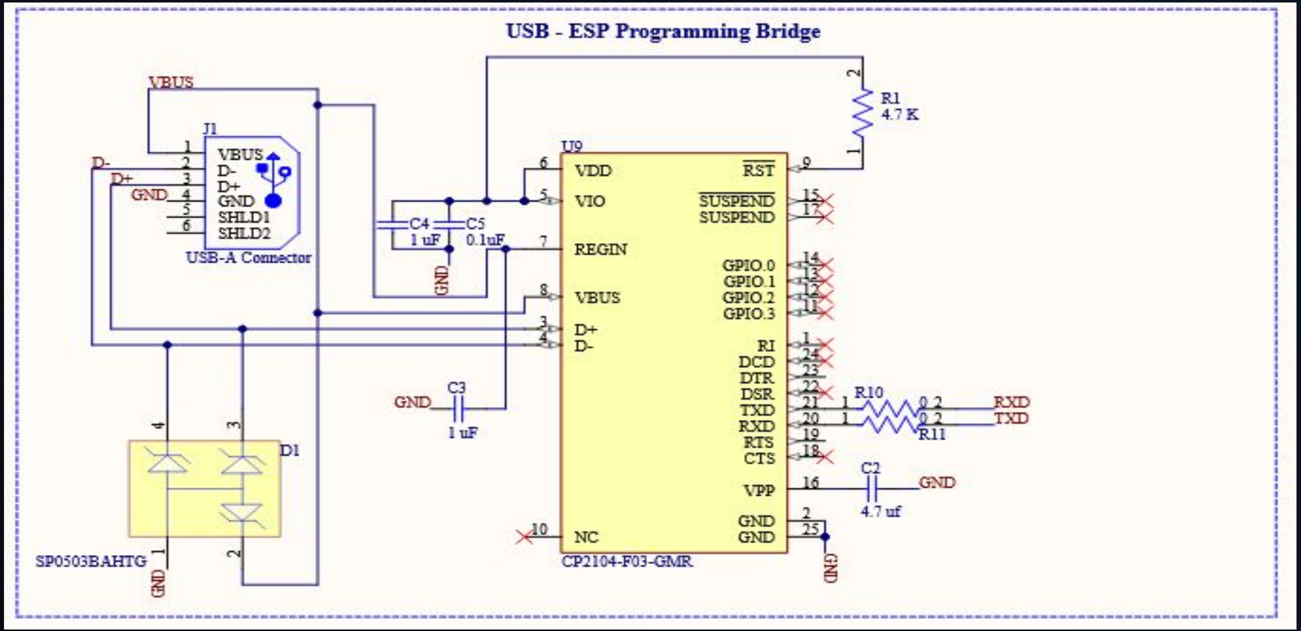
USB to UART Bridge

USB to UART Bridge



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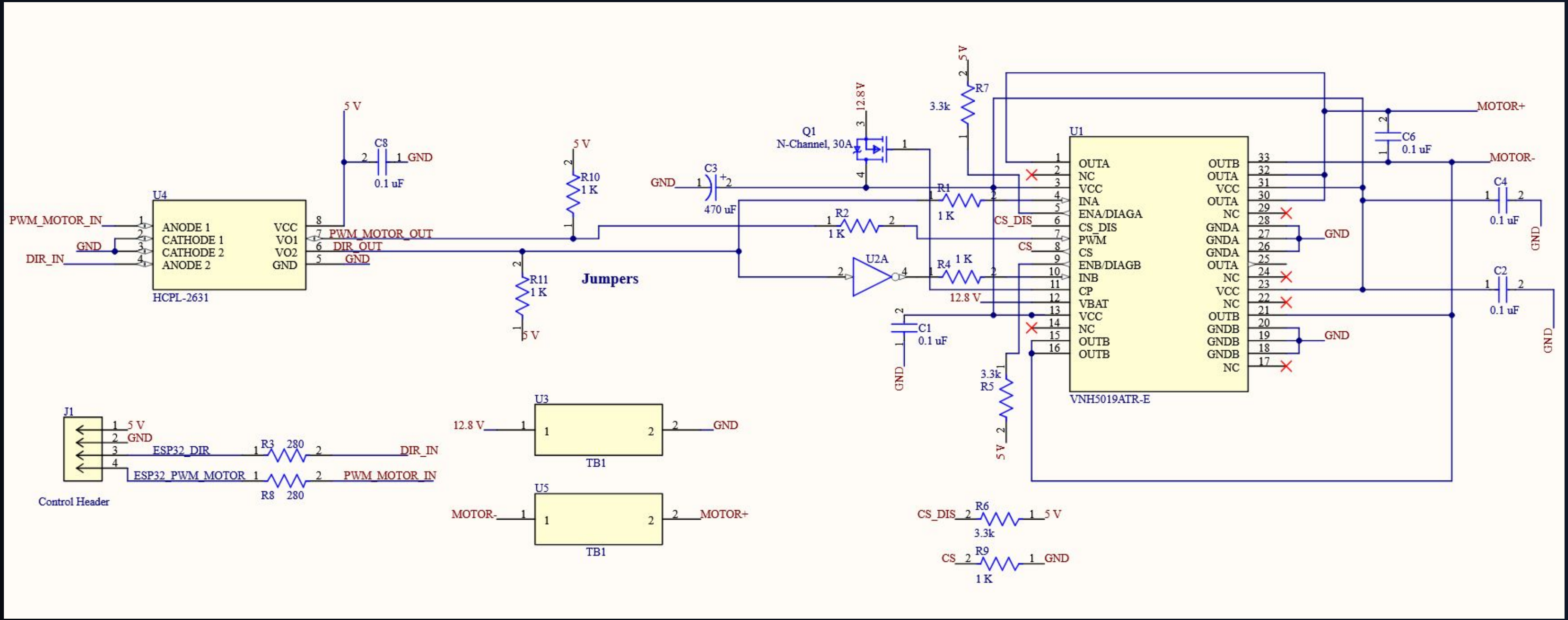
30 A Motor Driver

Motor Driver



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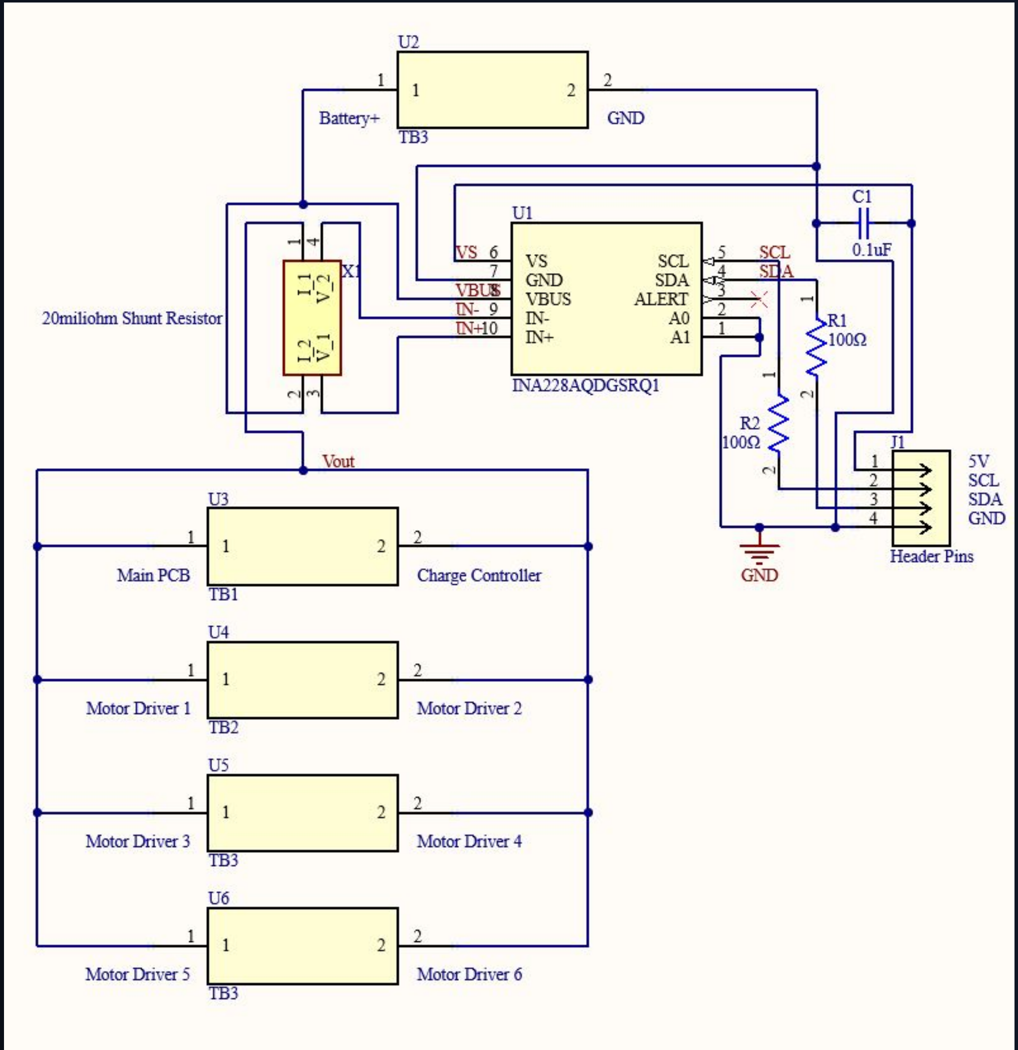
Energy Monitor



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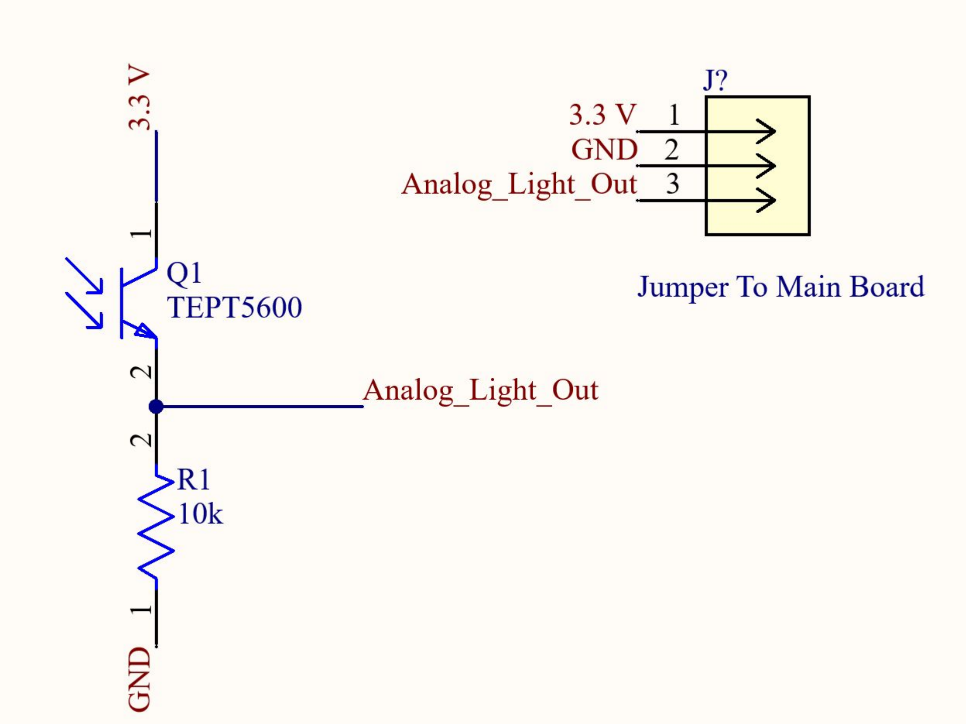
Light Sensor



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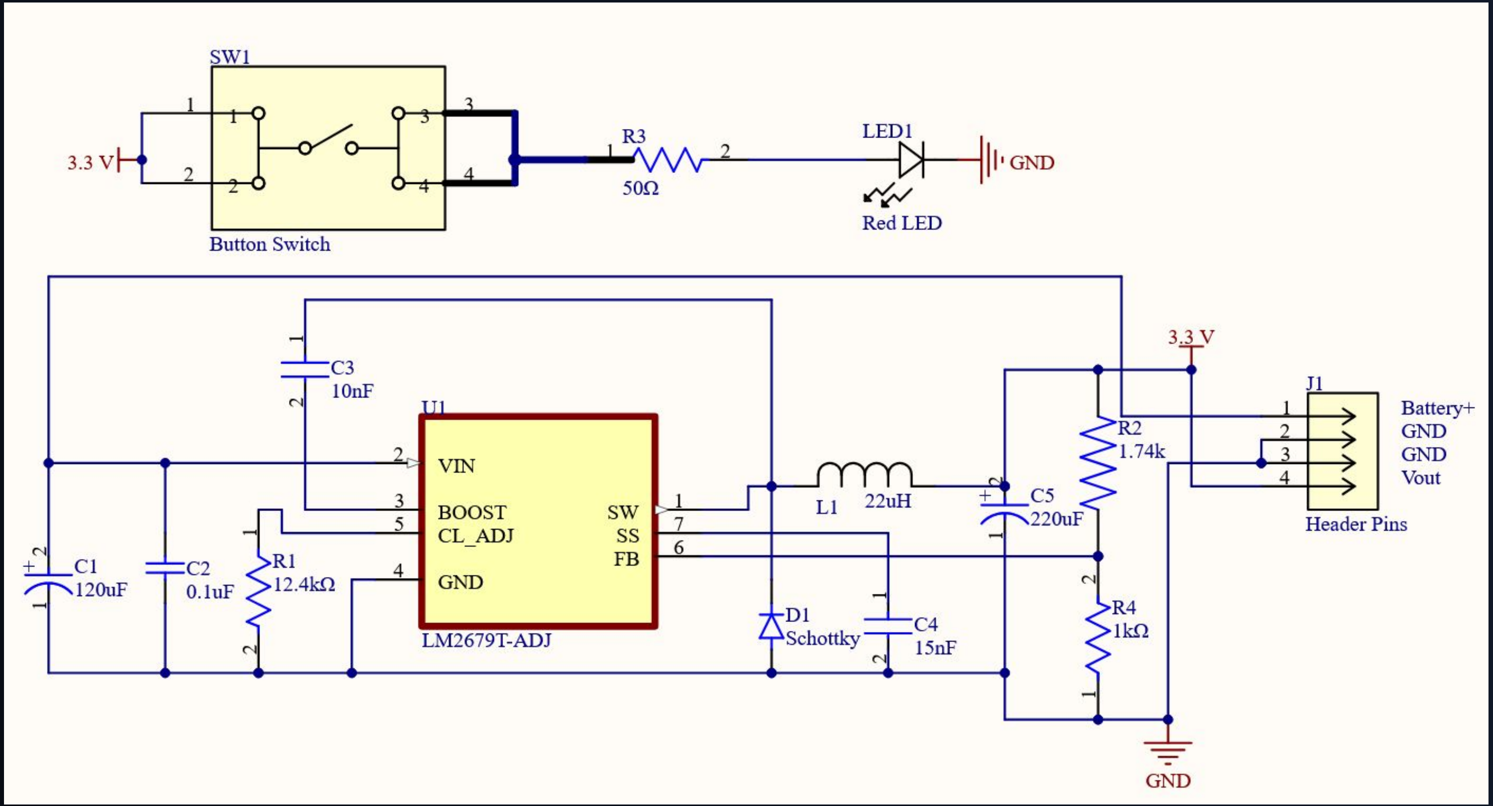
Voltage Regulator

3.3 V



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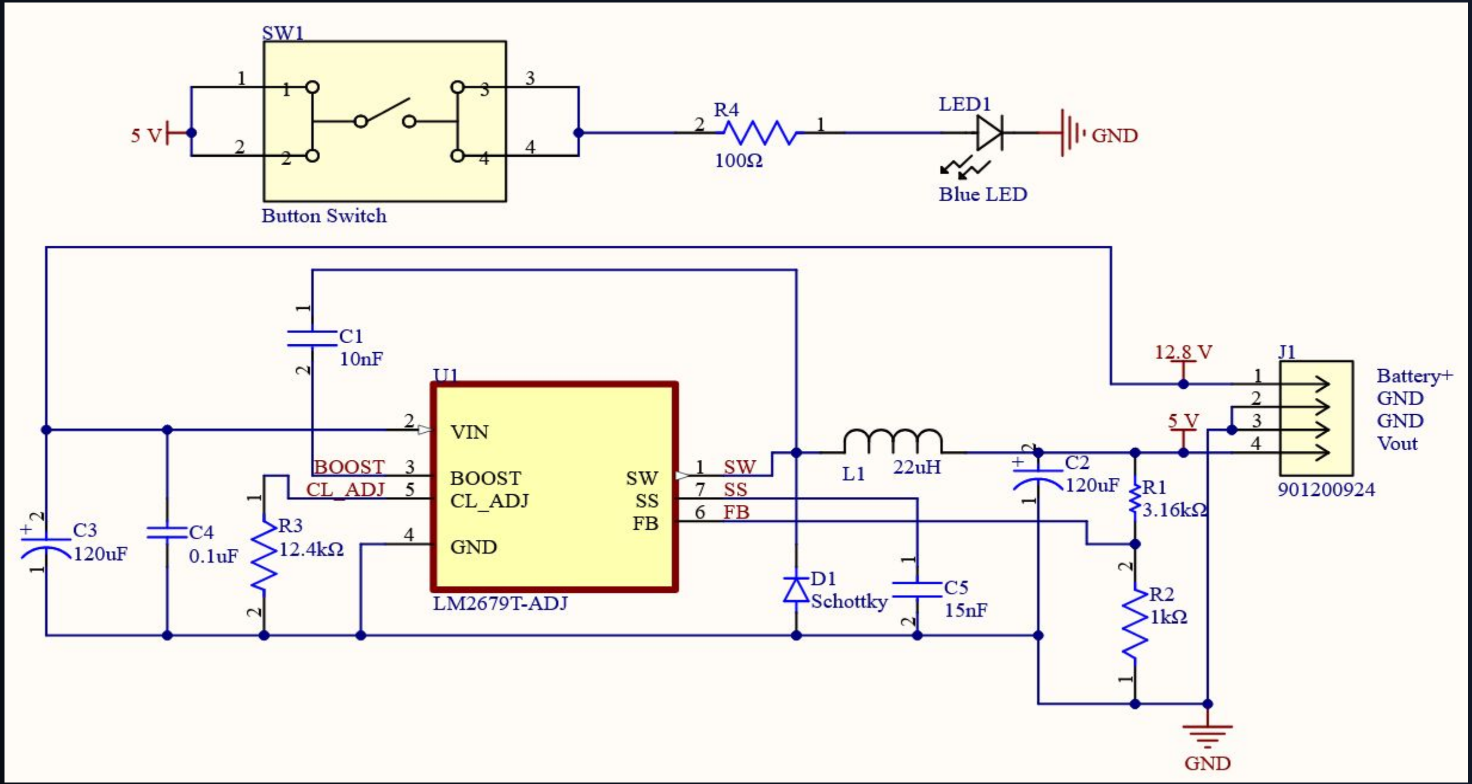
Voltage Regulator

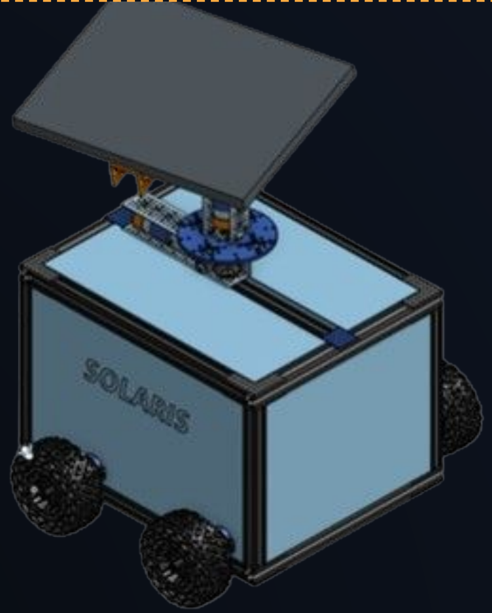
5.0 V



OWNER : RAFAEL

Mechanical Design

Robot Chassis Rendering



SELECTION

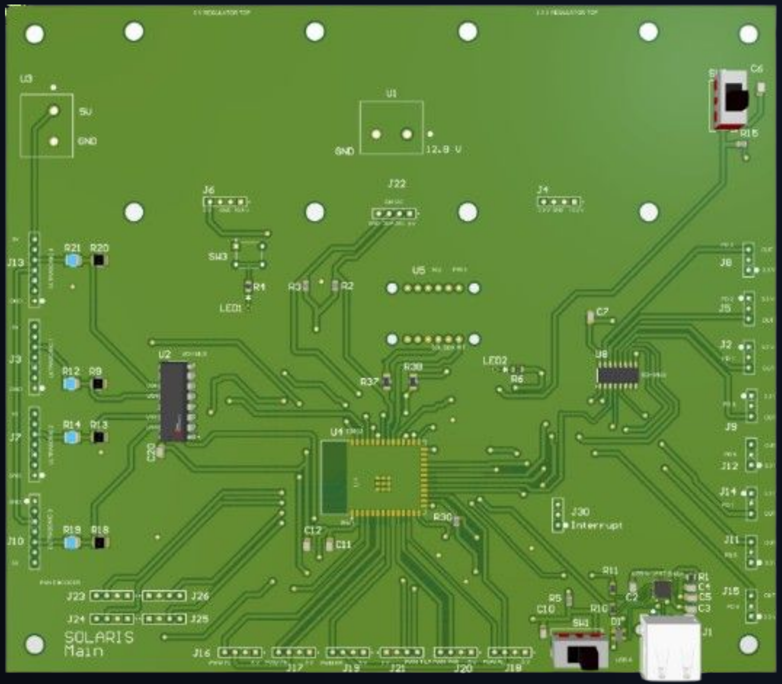
Mechanical Design

- **Dual-Axis Solar Mount System -** Precise tracking of sun throughout day.
- **All-Terrain Wheels -** High mobility of robot on outdoor terrain.
- **Multi-Purposed T-Slot Rails -** Easy to attach modular components anywhere on chassis.

OWNER : RAFAEL

PCB Layouts

Main PCB



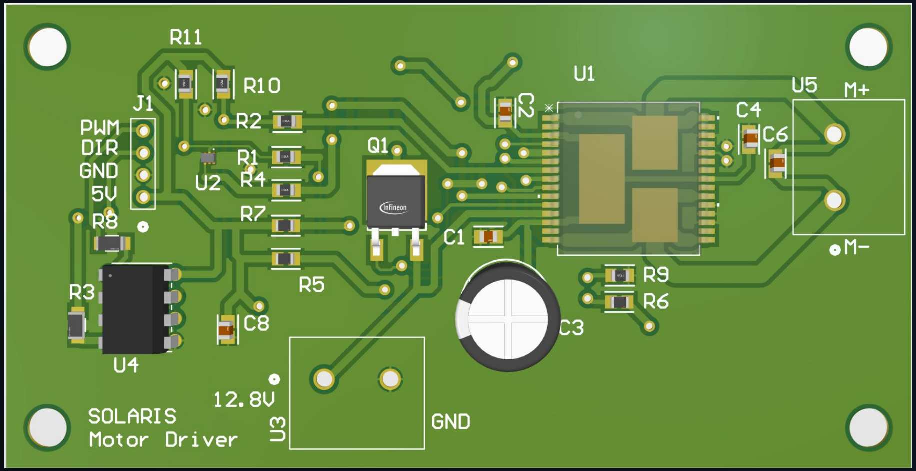
SELECTION

PCB layouts

- Ensuring every peripheral could be routed to
- Ensuring all bypass capacitors were as close to power pins as possible
- Ensuring I2C lines were as short as possible while minimizing trace crossover

OWNER : NATHAN

PCBs Part 2



Motor Driver

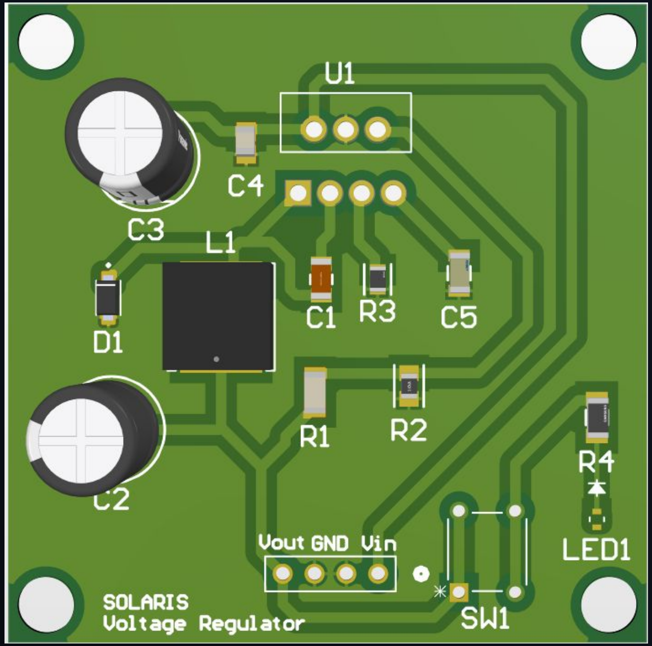
SELECTION

PCB layouts

- Ensuring traces for power lines were thick enough using polygon pours
- Placing capacitors for bypass on each power line as close as possible to the chip
- Dual-Channel Optocoupler selected and placed to ensure as much electrical isolation as possible

OWNER : NATHAN

PCBs Part 1



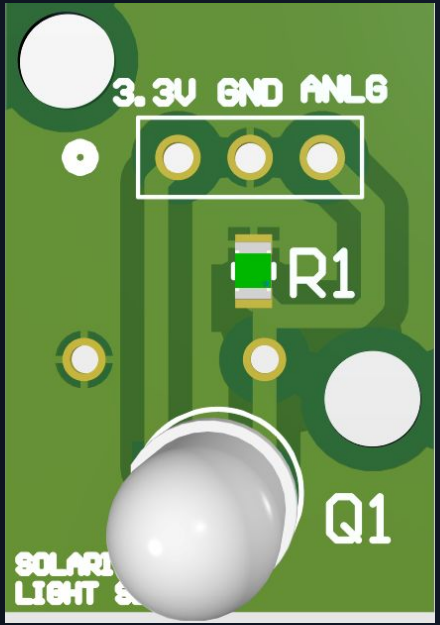
Voltage Regulator

OWNER : RAFAEL

SELECTION

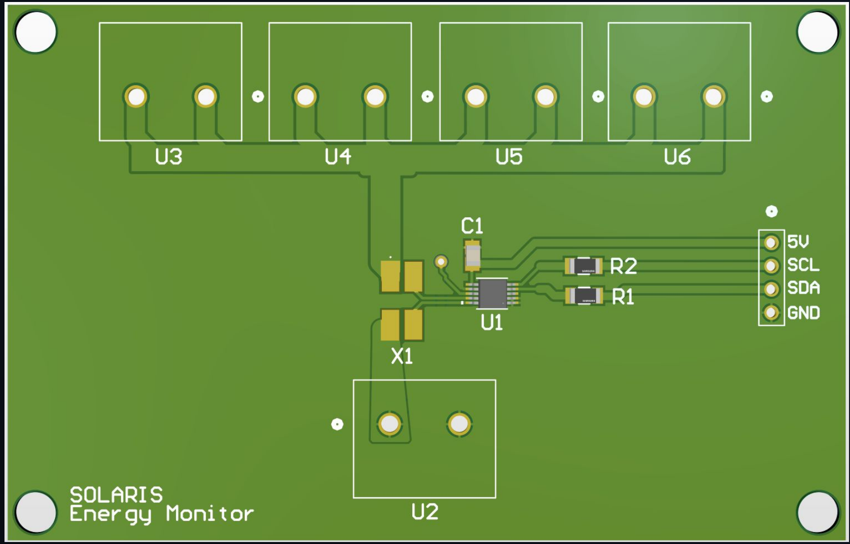
PCB layouts

- Utilized TI WEBENCH for voltage regulator output.
- Sized components on voltage regulators for up to 2.9 amps.
- Light Sensor is small sized to be fitted on corners of chassis as well as on solar panel.



Light Sensor

PCB Layouts



Energy Monitor

SELECTION

PCB layouts

- Bottleneck of system power. Rated for 5.3 amps.
- Voltage sensing lines kept short to minimize noise.
- Connectors on edges of board for functionality purposes.

OWNER : RAFAEL

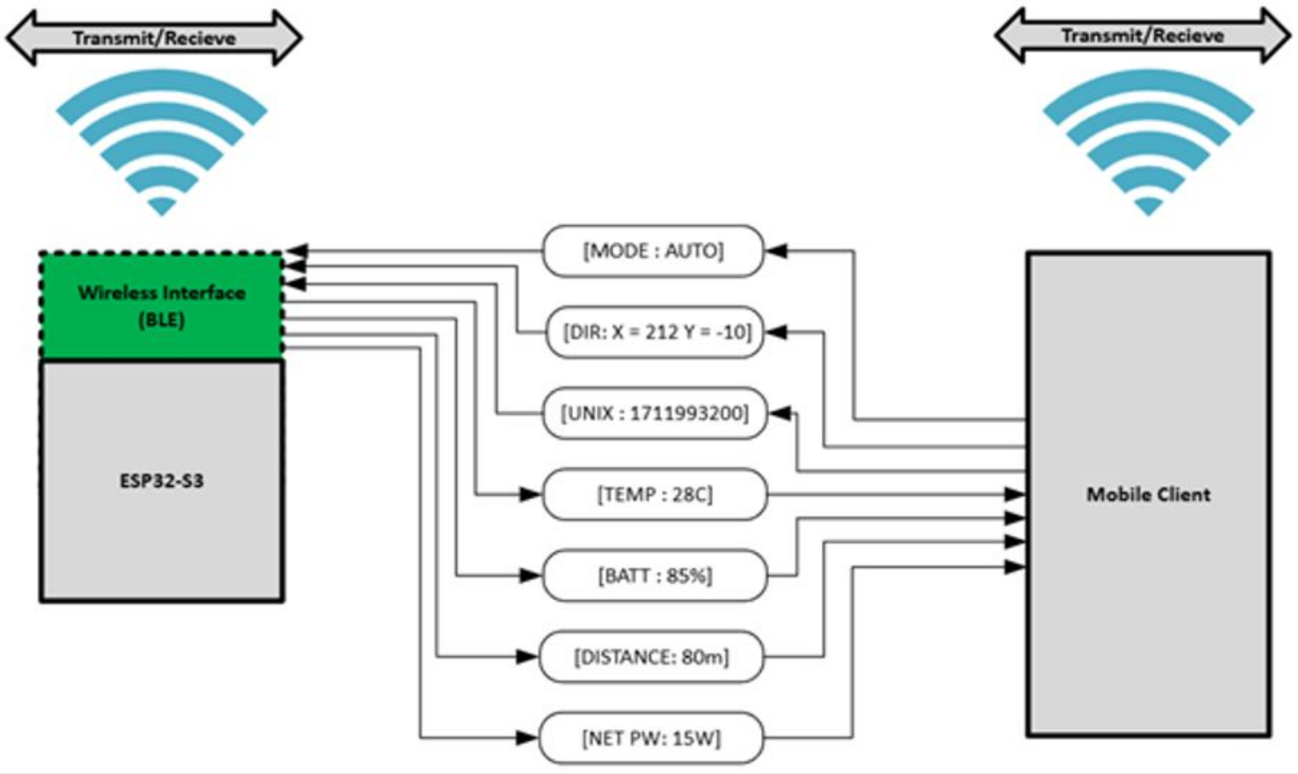
Software Backbone

Case Category	Specific Condition (Input)	Resulting State	Hardware Behavior
I. Connectivity	No Bluetooth paired	Search/Advertising	ESP32 LED blinks awaiting connection
	Bluetooth paired	Standby	Await Mode command (Auto/Manual)
II. Control Mode	Mode = Manual	Manual	Listen to controls; Service BLE with telemetry
	Mode = Stationary	Hibernation	Go to sleep in this spot
	Mode = Auto	Autonomous	Move to Autonomous case
III. Autonomous	Time = Night	Hibernation	Enter Deep Sleep
	Time = Day & Obstacle < 20cm	Obstacle Avoidance	Stop; Pivot using IMU and ultrasonics sensors for clear path
	Time = Day & Obstacle > 20cm	Solar Search	Move to Goal Evaluation case
IV. Goal Evaluation	Light < Threshold	Wandering	Moving towards luminous zone
	Light > Threshold	Solar Alignment	Stop wheels; Activate Pan/Tilt motors
	Alignment = Success	Hibernation	Shutdown to charge; Set wake timer

OWNER : GARRETT

Software Backbone (continued)

V. Safety System	Battery < 10%	Critical Stop	Send alert to App; kill all motors
	Robot is unable to move	Critical Stop	Send alert to App; kill all motors



OWNER : GARRETT

Project Status

Parts & Acquisition	99%
Design	99%
Prototyping	40%
Coding	60%
Overall	74.5%

OWNER : RYAN


SOLARIS
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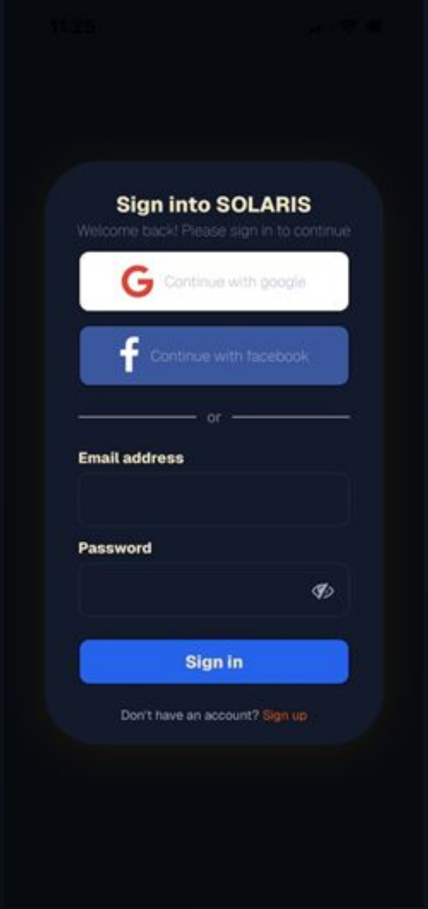
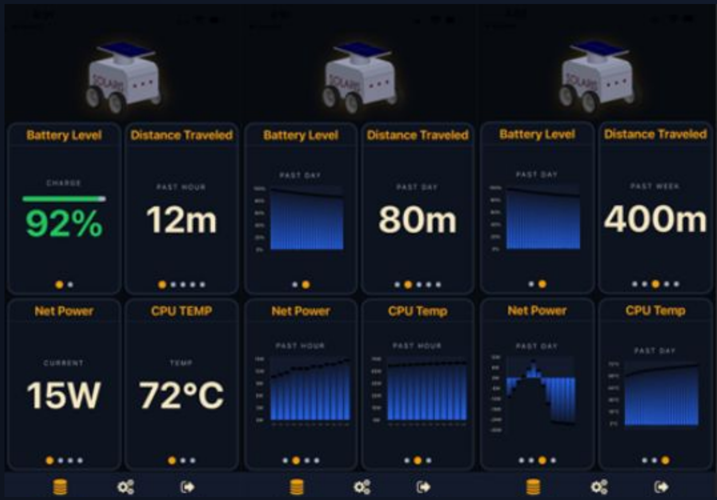
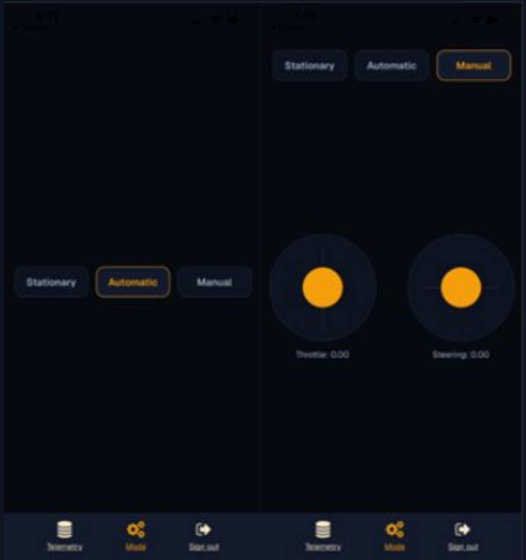
Sign Up

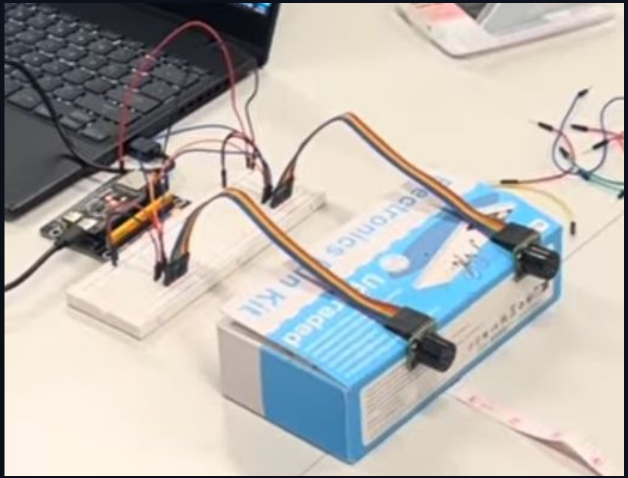

Sign into SOLARIS
Welcome back! Please sign in to continue
Continue with google
Continue with facebook
or
Email address
Password
Sign in
Don't have an account? Sign up


Battery Level
92%
Distance Traveled
12m
Battery Level
Distance Traveled
80m
Battery Level
Distance Traveled
400m
Net Power
15W
CPU TEMP
72°C
Net Power
CPU Temp
Net Power
CPU Temp

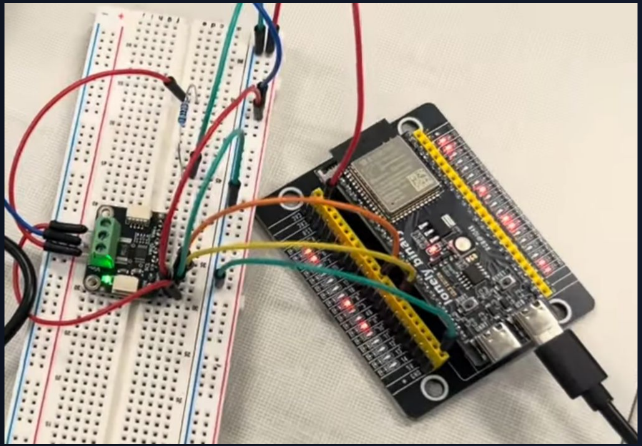

Stationary
Automatic
Manual
Stationary
Automatic
Manual

OWNER : RYAN

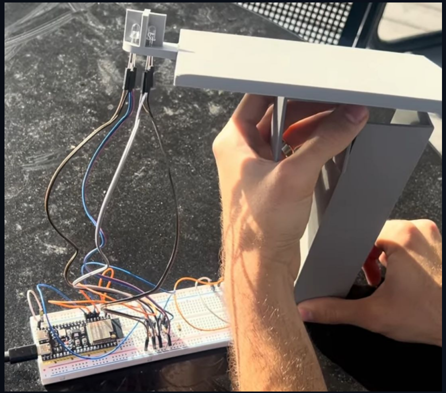
Ultrasonic Testing



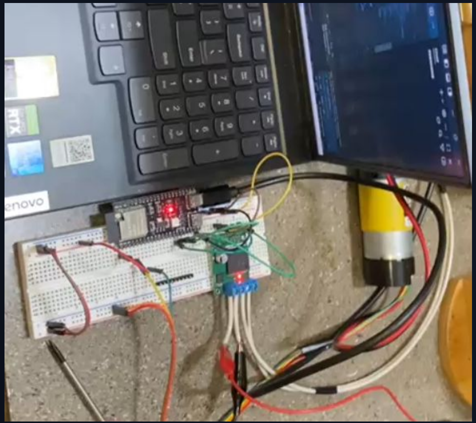
Energy Monitoring Unit Testing



Light Sensor Testing



Motor + Encoder Testing



OWNER : GARRET

Administrative Content

Item	Qty	Price
ESP32-S3 MCU	1	$6.56
DC Drive Motor	4	$227.96
Pan/Tilt Motor	2	$109.98
Motor Driver	6	$55.32
LiFePo4 Battery	1	$89.99
Newpowa Solar Power	1	$37.99
MPPT	1	$99.07

Item	Qty	Price
TEPT Phototransistor	8	$5.6
Energy Monitoring Unit	1	$5.27
ICM-20948 IMU	1	$19.95
PCB Fabrication	N/A	$230.07
Chassis	N/A	$623.22
Misc. Components	N/A	$427.48
Apple Dev Key	1	$99

Item	Qty	Price
Ultrasonic Sensor	4	$79.80
TOTAL		$2117.26

OWNER : GARRET

End of Presentation